Pattern-Based Programming Questions (All 34 Questions - Interview Style)

🔷 Square, Rectangle, and Triangle Patterns (1–15)

1. **Solid Square Pattern**
   Problem: Print a solid square of stars of size n.
   Input: n = 4
   Output:

```
* * * *
* * * *
* * * *
* * * *
```

```
n = 4

for i in range(n):
  for j in range(n):
    print("*", end=" ")
  print()
```

OUTPUT:

```
* * * *

* * * *

* * * *

* * * *
```

2. **Solid Rectangle Pattern**

Problem: Print a solid rectangle of m rows and n columns.
Input: m = 3, n = 5
Output:

```
* * * * *
* * * * *
* * * * *
```

```python
m = 3

n = 5


for i in range(m):

   for j in range(n):

      print("*", end=" ")

   print()
```

OUTPUT:

```
* * * * *

* * * * *

* * * * *
```

3. **Right-Angled Triangle (Left-Aligned)**

Problem: Print a left-aligned right-angled triangle.
Input: n = 5
Output:

```
*
* *
* * *
* * * *
* * * * *
```

```python
n = 5


for i in range(1, n + 1):

   for j in range(i):

      print("*", end=" ")
```

```
    print()
```

OUTPUT:

```
*
* *
* * *
* * * *
* * * * *
```

4. **Right-Angled Triangle (Right-Aligned)**

Input: n = 5

Output:

```
    *
   * *
  * * *
 * * * *
* * * * *
```

```
n = 5

for i in range(1, n + 1):
    print("  " * (n - i) + "* " * i)
```

OUTPUT:

```
    *
   * *
```

```
  * * *
 * * * *
* * * * *
```

5. **Inverted Triangle (Left-Aligned)**
Input: n = 5
Output:

```
* * * * *
* * * *
* * *
* *
*
```

```
n = 5

for i in range(n, 0, -1):
   print("* " * i)
```

OUTPUT:

```
* * * * *
* * * *
* * *
* *
*
```

6. **Inverted Triangle (Right-Aligned)**
Input: n = 5
Output:

```
* * * * *
 * * * *
  * * *
   * *
    *
```

```
n = 5

for i in range(n, 0, -1):
    print("  " * (n - i) + "* " * i)
```

OUTPUT:

```
* * * * *
 * * * *
  * * *
   * *
    *
```

7. **Centered Pyramid Pattern**

Input: n = 4

Output:

```
   *
  * * *
 * * * * *
* * * * * * *
```

```
n = 4

for i in range(1, n + 1):
    print("  " * (n - i) + "* " * (2 * i - 1))
```

OUTPUT:

```
   *
  * * *
 * * * * *
* * * * * * *
```

8. **Diamond Pattern**

Input: n = 3

Output:

```
  *
 * * *
* * * * *
 * * *
  *
```

```
n = 3

for i in range(1, n + 1):
    print("  " * (n - i) + "* " * (2 * i - 1))

for i in range(n - 1, 0, -1):
    print("  " * (n - i) + "* " * (2 * i - 1))
```

OUTPUT:

```
  *
```

```
 * * *
* * * * *
 * * *
  *
```

9. **Butterfly Pattern**
Input: n = 4
Output:

```
*     *
* *  * *
* * * * *
* *  * *
*     *
```

```
n = 4


for i in range(1, n + 1):
   print("* " * i + "  " * (2 * (n - i)) + "* " * i)


for i in range(n - 1, 0, -1):
   print("* " * i + "  " * (2 * (n - i)) + "* " * i)
```

OUTPUT:

```
*           *
* *       * *
* * *   * * *
* * * * * * * *
* * *   * * *
* *       * *
```

*   *

10. **Left-Aligned Half Diamond**

Input: n = 4

Output:

```
*
* *
* * *
* * * *
* * *
* *
*
```

```
n = 4


for i in range(1, n + 1):
    print("* " * i)


for i in range(n - 1, 0, -1):
    print("* " * i)
```

OUTPUT:

```
*
* *
* * *
* * * *
* * *
* *
*
```

11. **Right-Aligned Half Diamond**

Input: n = 4

Output:

```
   *
  * *
 * * *
* * * *
 * * *
  * *
   *
```

```
n = 4


for i in range(1, n + 1):
    print("  " * (n - i) + "* " * i)


for i in range(n - 1, 0, -1):
    print("  " * (n - i) + "* " * i)
```

OUTPUT:

```
   *
  * *
 * * *
* * * *
 * * *
  * *
   *
```

12. **Sandglass Pattern**

Input: n = 4

Output:

```
* * * *
 * * *
  * *
   *
  * *
 * * *
* * * *
```

```
n = 4


for i in range(n, 0, -1):
    print("  " * (n - i) + "* " * i)


for i in range(2, n + 1):
    print("  " * (n - i) + "* " * i)
```

OUTPUT:

```
* * * *
 * * *
  * *
   *
  * *
 * * *
* * * *
```

13. **Increasing Width Triangle**
Input: n = 5
Output:

```
*
* *
* * *
* * * *
* * * * *
```

```
n = 5


for i in range(1, n + 1):
   print("* " * i)
```

OUTPUT:

```
*

* *

* * *

* * * *

* * * * *
```

14. **Decreasing Width Triangle**
Input: n = 5
Output:

```
* * * * *
* * * *
* * *
* *
*
```

```
n = 5
```

```
for i in range(n, 0, -1):
   print("* " * i)
```

OUTPUT:

```
* * * * *
* * * *
* * *
* *
*
```

15. **Right-Aligned Hill Pattern**

Input: n = 4

Output:

```
   *
  * *
 * * *
* * * *
```

```
n = 4

for i in range(1, n + 1):
   print("  " * (n - i) + "* " * i)
```

OUTPUT:

```
   *
  * *
```

```
 * * *
* * * *
```

☐ Hollow Patterns (16–25)

16. **Hollow Square Pattern**
Problem: Print a hollow square of stars of size n.
Input: n = 4
Output:

```
* * * *
*     *
*     *
* * * *
```

```
n = 4


for i in range(n):

   for j in range(n):

      if i == 0 or i == n - 1 or j == 0 or j == n - 1:

         print("*", end=" ")

      else:

         print(" ", end=" ")

   print()
```

OUTPUT:

```
* * * *
*     *
*     *
* * * *
```

17. **Hollow Rectangle Pattern**

Problem: Print a hollow rectangle of m rows and n columns.

Input: m = 4, n = 5

Output:

```
* * * * *
*     *
*     *
* * * * *
```

```
m = 4

n = 5


for i in range(m):

   for j in range(n):

      if i == 0 or i == m - 1 or j == 0 or j == n - 1:

         print("*", end=" ")

      else:

         print(" ", end=" ")

   print()
```

OUTPUT:

```
* * * * *

*     *

*     *

* * * * *
```

18. **Hollow Right-Angled Triangle (Left-Aligned)**

Input: n = 5

Output:

```
*
* *
*   *
*     *
* * * * *
```

```
n = 5

for i in range(1, n + 1):
    for j in range(1, i + 1):
        if j == 1 or j == i or i == n:
            print("*", end=" ")
        else:
            print(" ", end=" ")
    print()
```

OUTPUT:

```
*
* *
*   *
*     *
* * * * *
```

19. **Hollow Right-Angled Triangle (Right-Aligned)**

Input: n = 5

Output:

```
    *
   * *
  *   *
 *     *
* * * * *
```

```
n = 5

for i in range(1, n + 1):
    print("  " * (n - i), end="")

    for j in range(1, i + 1):
        if j == 1 or j == i or i == n:
            print("* ", end="")
        else:
            print("  ", end="")

    print()
```

OUTPUT:

```
    *
   * *
  *   *
 *     *
* * * * *
```

20. **Hollow Inverted Triangle (Left-Aligned)**

Input: n = 5

Output:

```
* * * * *
*     *
*   *
* *
*
```

```
n = 5

for i in range(n, 0, -1):
    for j in range(1, i + 1):
        if j == 1 or j == i or i == n:
            print("*", end=" ")
        else:
            print(" ", end=" ")
    print()
```

OUTPUT:

```
* * * * *
*     *
*   *
* *
*
```

21. **Hollow Inverted Triangle (Right-Aligned)**

Input: n = 5

Output:

```
* * * * *
 *    *
  *  *
   * *
    *
```

```
n = 5


for i in range(n, 0, -1):
   print("  " * (n - i), end="")


   for j in range(1, i + 1):
      if j == 1 or j == i or i == n:
         print("* ", end="")
      else:
         print("  ", end="")


   print()
```

OUTPUT:

```
* * * * *
 *    *
  *  *
   * *
    *
```

22. **Hollow Pyramid Pattern**

Input: n = 4

Output:

```
   *
  * *
 *   *
* * * * * * *
```

```
n = 4

for i in range(1, n + 1):
   print("  " * (n - i), end="")

   for j in range(1, 2 * i):
      if j == 1 or j == 2 * i - 1 or i == n:
         print("* ", end="")
      else:
         print("  ", end="")

   print()
```

OUTPUT:

```
   *
  * *
 *   *
* * * * * * *
```

23. **Hollow Diamond Pattern**

Input: n = 3

Output:

```
  *
 * *
*   *
 * *
  *
```

```
n = 3

for i in range(1, n + 1):
    print("  " * (n - i), end="")

    for j in range(1, 2 * i):
        if j == 1 or j == 2 * i - 1:
            print("* ", end="")
        else:
            print("  ", end="")

    print()

for i in range(n - 1, 0, -1):
    print("  " * (n - i), end="")

    for j in range(1, 2 * i):
        if j == 1 or j == 2 * i - 1:
            print("* ", end="")
        else:
```

```
        print("  ", end="")

    print()
```

OUTPUT:

```
  *
 *  *
*     *
 *  *
  *
```

24. **Hollow Butterfly Pattern**

Input: n = 4

Output:

```
*     *
* *  * *
*  *  *
*     *
*  *  *
* *  * *
*     *
```

```
n = 4

for i in range(1, n + 1):

    for j in range(1, i + 1):

        if j == 1 or j == i:
```

```
        print("*", end=" ")
      else:
        print(" ", end=" ")

  print("  " * (2 * (n - i)), end="")

  for j in range(1, i + 1):
    if j == 1 or j == i:
      print("*", end=" ")
    else:
      print(" ", end=" ")

  print()

for i in range(n - 1, 0, -1):
  for j in range(1, i + 1):
    if j == 1 or j == i:
      print("*", end=" ")
    else:
      print(" ", end=" ")

  print("  " * (2 * (n - i)), end="")

  for j in range(1, i + 1):
    if j == 1 or j == i:
      print("*", end=" ")
    else:
      print(" ", end=" ")
```

```
    print()
```

OUTPUT:

```
*         *
**       **
*  *   *  *
*   **   *
*   **   *
*  *   *  *
**       **
*         *
```

25. **Hollow Hourglass Pattern**
Input: n = 5
Output:

```
* * * * *
*     *
 *  *
   *
 *  *
*     *
* * * * *
```

n = 5

```
# Top
print("* " * n)

# Upper hollow part
for i in range(1, n - 2):
    print("  " * i, end="")

    for j in range(n - i):
        if j == 0 or j == n - i - 1:
            print("* ", end="")
        else:
            print("  ", end="")

    print()

# Middle
print("  " * 2 + "*")

# Lower hollow part
for i in range(n - 3, 0, -1):
    print("  " * i, end="")

    for j in range(n - i):
        if j == 0 or j == n - i - 1:
            print("* ", end="")
        else:
            print("  ", end="")
```

    print()

Bottom

print("* " * n)

OUTPUT:

```
* * * * *
*     *
 *  *
  *
 *  *
*     *
* * * * *
```

🔢 Number-Based Patterns (26–34)

26. **Increasing Number Triangle**
Problem: Print numbers from 1 to n in triangle form.
Input: n = 5
Output:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

n = 5

for i in range(1, n + 1):

```python
    for j in range(1, i + 1):
        print(j, end=" ")
    print()
```

OUTPUT:

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

27. **Repeating Row Number Triangle**

Input: n = 5
Output:

1
2 2
3 3 3
4 4 4 4
5 5 5 5 5

```python
n = 5

for i in range(1, n + 1):
    for j in range(i):
        print(i, end=" ")
    print()
```

OUTPUT:

1

2 2

3 3 3

4 4 4 4

5 5 5 5 5

28. **Continuous Number Triangle**

Input: n = 4

Output:

1
2 3
4 5 6
7 8 9 10

```
n = 4

num = 1


for i in range(1, n + 1):

   for j in range(i):

      print(num, end=" ")

      num += 1

   print()
```

OUTPUT:

1

2 3

4 5 6

7 8 9 10

29. **Reverse Row Number Triangle**

Input: n = 5

Output:

1
2 1
3 2 1
4 3 2 1
5 4 3 2 1

```
n = 5

for i in range(1, n + 1):
  for j in range(i, 0, -1):
    print(j, end=" ")
  print()
```

OUTPUT:

1

2 1

3 2 1

4 3 2 1

5 4 3 2 1

30. **Inverted Number Triangle**

Input: n = 5

Output:

5 4 3 2 1
4 3 2 1
3 2 1
2 1
1

```
n = 5

for i in range(n, 0, -1):
   for j in range(i, 0, -1):
      print(j, end=" ")
   print()
```

OUTPUT:

5 4 3 2 1

4 3 2 1

3 2 1

2 1

1

31. **Right-Aligned Number Triangle**

Input: n = 5

Output:

```
    1
   1 2
  1 2 3
```

```
 1 2 3 4
1 2 3 4 5
```

```
n = 5

for i in range(1, n + 1):
   print("  " * (n - i), end="")

   for j in range(1, i + 1):
      print(j, end=" ")

   print()
```

OUTPUT:

```
    1
   1 2
  1 2 3
 1 2 3 4
1 2 3 4 5
```

32. **Pyramid Number Pattern**

Input: n = 4

Output:

```
   1
  1 2 1
 1 2 3 2 1
1 2 3 4 3 2 1
```

```
n = 4

for i in range(1, n + 1):
    print("  " * (n - i), end="")

    for j in range(1, i + 1):
        print(j, end=" ")

    for j in range(i - 1, 0, -1):
        print(j, end=" ")

    print()
```

OUTPUT:

```
      1
    1 2 1
  1 2 3 2 1
1 2 3 4 3 2 1
```

33. **Even Number Triangle**

Input: n = 5

Output:

```
2
2 4
2 4 6
2 4 6 8
2 4 6 8 10
```

```python
n = 5

for i in range(1, n + 1):
    for j in range(1, i + 1):
        print(2 * j, end=" ")
    print()
```

OUTPUT:

```
2
2 4
2 4 6
2 4 6 8
2 4 6 8 10
```

34. **Odd Number Triangle**

Input: n = 5

Output:

```
1
1 3
1 3 5
1 3 5 7
1 3 5 7 9
```

```python
n = 5

for i in range(1, n + 1):
    for j in range(1, i + 1):
        print(2 * j - 1, end=" ")
```

```
    print()
```

OUTPUT:

```
1
1 3
1 3 5
1 3 5 7
1 3 5 7 9
```